

CoinWise AI — Advanced Fintech Synthesis

AI-driven, alternative-data, VND-localized learn-to-trade platform for the Vietnamese market

Project Report · **AI Integration & API Refinement** | Prepared for: UII Incubator — Demo Day | Stage: deployed MVP (paper-trading)

1. Executive Summary & Product Concept

CoinWise AI is a learn-by-doing investing platform for Vietnam's retail crypto market. Crypto adoption here is among the highest in the world, yet retail tooling is English-first, USD-first, and offers generic price charts with no localized decision support — a large audience that is highly active but under-educated and over-exposed to FOMO and scams. We close that gap by combining four things into one product:

- **(1) Academy** — structured courses on trading and on how to use AI in crypto;
- **(2) Risk-free paper trading** — every user gets 1,000,000 USDT of virtual capital to practise real strategies with zero downside;
- **(3) An AI copilot + alternative-data intelligence engine** — proprietary signals and an agentic chatbot that explain *why* to act;
- **(4) Arena** — skill-based trading tournaments where users pay a VND entry fee and win a VND prize pool, turning learning into an engaging, social loop.

The thesis: people do not learn to trade by reading — they learn by practising safely, with feedback. CoinWise AI lets users practise against the real market, get AI-graded feedback and signals, and compete — so they build genuine skill before risking real money. Underneath this consumer wrapper sits the graded technical core of this assignment: a **proprietary alternative-data AI engine**, a **custom VND-localized OpenAPI server** that fixes the currency gaps global providers (Stripe/Plaid) leave open, and an **agentic Gemini assistant** that executes account actions through natural language. These three pillars are what make the business defensible and incubation-ready — and they are all live in the deployed MVP today.

1.1 The problem and the Vietnamese opportunity

Vietnam is one of the world's highest crypto-adoption markets (consistently top-ranked in Chainalysis' Global Adoption Index), with ~100M people, a mobile-first median age of ~33, and an estimated 17–20M crypto holders. Yet retail investors face three structural gaps: tooling is **not localized** (USD/English-first; Stripe/Plaid don't handle VND or local banking natively); intelligence is **generic** (charts, but no “should I act, and why”); and adoption is high while **literacy is low** (heavy exposure to scams, FOMO and over-leverage). Our wedge is to be Vietnamese-first, VND-first and AI-first — teaching users how to use AI in crypto while they practise risk-free.

Sizing (assumption — to validate)	Definition	Estimate
TAM	VN adults using crypto / online investing	~17–20M users
SAM	Mobile-first 18–40, urban, willing to pay for tools	~3–4M users
SOM (3-yr)	Realistic capture with a localized AI product	~150k–300k (target)

2. AI Methodology — Alternative Data (Part A · 30%)

Alternative data is any non-traditional signal beyond price and transaction history — here, public social/news text, community voting, and market-psychology indices. Retail traders react to narrative and crowd mood long before it shows up on the chart; capturing that early, and explaining it, is our edge. The **Alt-Data Lab** in the app exposes the full chain transparently: *real data* → *AI analysis* → *a concrete fintech action*.

2.1 Data sources (live, free, no paid keys)

Source	What it provides	Role
Hacker News (Algolia)	Story titles + upvote scores mentioning a coin	Primary text corpus
Reddit (r/CryptoCurrency...)	Post titles/bodies + upvotes	Supplementary text
alternative.me Fear &	Market-wide psychology index 0–100	Fusion signal +

Greed		Social Pulse
CoinGecko community	Up/down vote %, dev & community score	Fusion signal

Production-honest by design: sources are called in parallel; if one is blocked (Reddit frequently IP-blocks with 403), the pipeline does not crash — it marks that stage *partial/failed*, re-allocates fusion weights to the surviving sources, and stamps every datapoint with provenance (LIVE / HYBRID / DEMO). Nothing is a black box.

2.2 The four-stage pipeline · GET /api/v1/ai/alt-data/pipeline/:symbol

Collect → **Analyse** → **Fuse** → **Apply**. Each post is normalised into a weighted document (weight = engagement, so high-upvote posts move the score more). Four AI techniques from the course then run on the same corpus:

- **VADER lexicon NLP** — rule-based valence scoring with negation, intensifier and ALL-CAPS handling; compound score in $[-1, +1]$. Fully *explainable per word* (the UI tags each token, e.g. “rally +2.3”, “crash −3.1”).
- **Multinomial Naive Bayes (trained by us)** — classifies each document as positive/negative/neutral with probabilities and decisive tokens. This is the “real” learned model (see 2.3).
- **Z-score anomaly detection** — a rolling baseline of mention volume; a spike of $>1.5\sigma$ flags a surge of retail attention (an early FOMO indicator).
- **Multi-source fusion** — text blend ($NB \times 0.6 + VADER \times 0.4$) combined with CoinGecko and Fear & Greed into a single composite score, confidence, and signal (STRONG_BUY ... STRONG_SELL), with automatic re-weighting when a source is down.

2.3 The trained model — distant supervision, “Python train, TypeScript serve”

Why Naive Bayes: a classic, transparent text classifier whose every prediction decomposes into per-token contributions — exactly what we need to *show* the reasoning, not just a number. It is trained in a reproducible Jupyter notebook using **distant supervision**: ~289 hand-labeled “gold” documents plus ~500 VADER-auto-labeled

“silver” documents scraped from Hacker News + RSS. Critically, the **test set is gold-only** (human-verified labels) and silver is used for training only — so accuracy cannot be inflated by the auto-labeller. Result of the model currently served: **accuracy $\approx$ 73.6%, macro-F1 $\approx$ 73.9%, vocabulary 2,268** (GET /api/v1/ai/alt-data/model/info).

The bridge: the notebook's final cell exports the learned weights (log-prior, per-token log-likelihoods, OOV term) into auto-generated TypeScript, and the runtime re-implements the exact Naive Bayes log-space computation. The model therefore trains in Python/scikit-learn (the academic deliverable) but **serves inside a serverless TypeScript function with no Python at runtime** — identical predictions, production-ready latency.

2.4 From signal to product — and why this is our competitive advantage

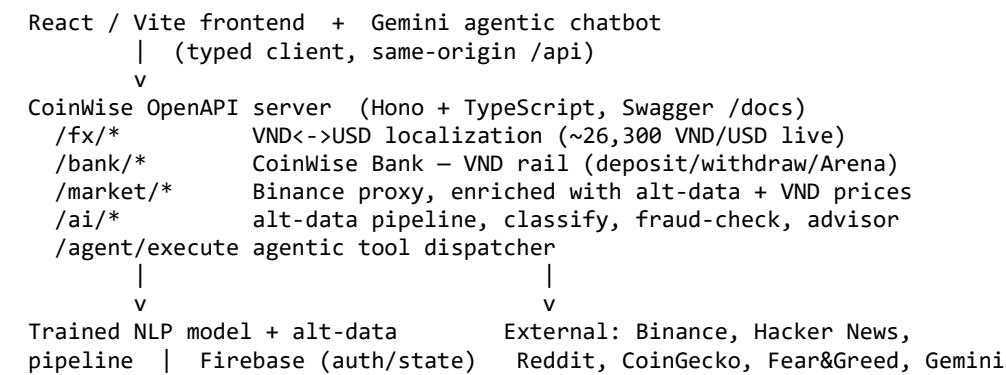
Alt data does not stop at a pretty number; it feeds two business modules:

- **Fraud Shield** — flags momentum-chasing / FOMO orders (e.g. buying into extreme-negative sentiment or right at an attention spike) and asks for confirmation, a consumer-protection feature for an inexperienced audience.
- **AI Advisor** — tilts a risk-profiled portfolio allocation by $\pm 25\%$ according to the strength and confidence of the composite signal.

The moat — a proprietary data flywheel. Our **Community Pulse** lets users post their take on any coin; each comment is scored live by *our own trained model* and aggregated into a Vietnam-specific community-sentiment signal that sharpens the AI verdict. The more users engage, the larger our first-party, locally-labeled dataset grows, and the better the model becomes — a compounding asset a global incumbent cannot buy off-the-shelf or replicate without our user base. Three structural advantages result: (1) **explainability** — every signal is defensible token-by-token; (2) **resilience** — graceful degradation and provenance keep it production-honest; and (3) **locality** — a dataset and signal tuned to the Vietnamese market that USD/English-first incumbents structurally lack.

3. Server Architecture — Custom OpenAPI Server (Part B · 30%)

The problem we fix: global sandboxes (Stripe, Plaid) do not process VND natively or integrate with local banking rails, and they cannot serve our AI analytics in a structured contract. So instead of bending a rigid foreign sandbox, we built our **own OpenAPI server** (Hono + TypeScript), documented with Swagger UI at **/docs** and a machine-readable **openapi.yaml**, deployed as a Vercel serverless function alongside the React/Vite frontend.



3.1 Solving real market friction — the VND money layer

- **FX endpoints** — `/fx/rates` and `/fx/convert` expose live USD↔VND (~26,300 VND/USD), so every balance, price and fee is correct in local currency, end to end.
- **CoinWise Bank** — a custom local banking rail (`/bank/...`) that processes deposits, withdrawals, subscription billing, and Arena entry fees & prize payouts *in VND*. This is precisely the “replace the global sandbox with a localized backend” the rubric rewards, and the money rail behind the Arena tournaments in our product concept.

3.2 API architecture — AI data straight to the frontend and the agent

Endpoint	Purpose
GET <code>/ai/alt-data/pipeline/:symbol</code>	Full alt-data → AI signal pipeline for a coin
POST <code>/ai/alt-data/classify</code>	Score any text with the trained model (live)
GET <code>/ai/alt-data/model/info</code>	Model metadata + evaluation metrics
POST <code>/ai/fraud-check-real</code>	Alt-data-driven transaction risk
POST <code>/agent/execute</code>	Tool dispatcher consumed by the AI agent
GET <code>/fx/rates</code> · <code>/bank/...</code>	VND localization + local banking rail

Technical defensibility: a real, documented, currency-correct contract with provenance stamping and graceful source degradation — not a thin wrapper over a foreign API. The server is the single source of truth for paper-account state, so the UI and the chatbot always agree.

4. Agentic AI & Localized UX (Part C · 20%)

Beyond chat to action. The assistant is a **Gemini function-calling agent**: it does not just answer questions, it calls our OpenAPI tools to actually do things. Registered tools include `getPortfolio`, `getSentiment`, `getAIInsight`, `convertCurrency`, and `placeTrade` (with a confirmation card). It understands localized intent — a user can type “*mua giúp tôi 5.000.000đ BTC*” (“*buy 5,000,000 VND of BTC*”), and the agent converts VND→USD, quotes the trade, asks for confirmation, then executes it through the server — every tool call shown transparently in-chat.

- **Localized transactions** — a global USD↔VND toggle re-renders all amounts via a currency context; balances, prices and fees are processed in VND.
- **Next-gen trading UX** — live candlestick charts (lightweight-charts) with exchange-grade indicators, an Alt-Data Lab, Social Pulse, AI Advisor, Fraud Shield, and the Arena — all wired to the same custom backend.

4.1 The learn-to-trade loop — Academy → Practice → Compete

This is the engagement engine of the business concept. New users start in the **Academy** (structured lessons on trading and, crucially, on how to use AI tools in crypto), then immediately **practise** those lessons against the live market with 1,000,000 USDT of virtual capital — zero financial risk, real price action. The AI copilot turns each session into a feedback loop: it explains the alt-data signal behind a move, flags FOMO entries via Fraud Shield, and suggests allocations via the AI Advisor, so users don't just trade — they learn *why*. Finally, the **Arena** converts skill into a social, repeatable habit: users pay a VND entry fee through CoinWise Bank, compete on a real-time leaderboard, and win a VND prize pool. The result is a virtuous loop — learn, practise, compete, improve — that

drives retention today and produces the labeled engagement data that compounds the moat for tomorrow.

5. Updated Roadmap & Incubation Readiness (Part D · 20%)

Why the AI makes the business incubation-viable for UII: (1) a **data moat** that compounds with usage; (2) **real localized infrastructure** no global API offers in VND; and (3) an **education-first, paper-trading on-ramp** that has a low regulatory surface today while de-risking a future regulated, real-money launch. Revenue lines are already surfaced in the product and activate as we move to real money:

Revenue line	Mechanism (built)	Pricing
Subscriptions	Free / Pro / Elite tiers	Pro \$19/mo (~500k ^d) · Elite \$99/mo
Arena	Paid skill tournaments, VND prize pools	Entry \$5–\$500 (rake on fees)
Earn / Yield	Staking products	APY 2.5%–18% (yield spread)
Future	Brokerage spread, B2B signal/data API	Transaction + data revenue

Commercial scaling roadmap

Phase	Milestones
0 — Now (done)	Deployed AI MVP: alt-data engine, VND OpenAPI server, agentic AI, data-flywheel foundation
1 — Incubation (0–6 mo)	eKYC/compliance design; validate conversion & ARPU; expand alt-data (on-chain, app-usage); harden infra
2 — Real-money beta (6–12 mo)	Integrate local banking rail (VietQR/Napas); licensed/partnered brokerage; paid subscriptions live
3 — Scale (12–24 mo)	Grow to SOM target; B2B data/signal API; new asset classes; SEA expansion

5.1 Illustrative unit economics & key risks

Unit economics (assumption — to validate): free→paid conversion of 3–5%; blended paid ARPU ~\$30/mo; CAC reduced by the built-in referral flywheel and organic, viral Community/Arena loops; target LTV/CAC > 3x within 18 months of real-money launch. Validating conversion and ARPU is an explicit incubation milestone, not a claimed result.

Risk	Mitigation
Regulatory uncertainty on digital assets in VN	Launch education + paper-trading (low regulatory surface); add eKYC/AML and SBV-aligned compliance before any real money
Free data-source reliability (e.g. Reddit 403)	Graceful degradation + provenance stamping today; add redundant and paid sources as we scale
Model accuracy (~74%) on noisy social text	Confidence thresholds, human-in-the-loop, and continuous retraining via the Community-Pulse data flywheel
Retention & monetization unproven	Engagement-first paper-trading validates value before real funds; conversion/ARPU tracked from day one

Demo Day flow (90 seconds)

Alt-Data Lab (scrape → VADER + trained NB → composite signal, with provenance + model metrics) → **apply** it in Fraud Shield / AI Advisor → **Community Pulse** (post a take, scored live — the data flywheel) → **agentic chatbot**: “buy 5,000,000^d BTC” → quote → confirm → executed in VND through the OpenAPI server → close on the moat and the incubation ask. Live MVP, Swagger docs at /docs, and the training notebook are all available to the panel.

How this maps to the evaluation rubric

Criterion (weight)	Evidence in CoinWise AI (built)
AI & Alternative Data (30%)	Live alt-data pipeline (Hacker News / Reddit / CoinGecko / Fear & Greed) + our own distant-supervision-trained Naive Bayes (acc ~73.6%) and VADER; signals drive Fraud Shield & AI Advisor; Community-Pulse data flywheel = the moat
Custom API Infrastructure	Proprietary Hono OpenAPI server with Swagger /docs; VND-native

(30%)	FX + CoinWise Bank rail; endpoints serving AI alt-data straight to the frontend and the agent
MVP UX & Agentic AI (20%)	Gemini function-calling agent executes trades & calls backend APIs via natural language (“buy 5,000,000 ^d BTC”); VND/USD toggle; live charts; learn-to-trade loop
Technical Documentation (20%)	This report + openapi.yaml/Swagger UI + reproducible training notebook + deployed link; incubation roadmap, compliance and unit economics for UII